

Guion para exposición oral – 20 minutos

Statistical Characterization and Optimization of Format Changeovers in DAMM's El Prat Bottling Lines

Jose Calatayud Mateu

Estructura recomendada. La exposición debe contar una historia:

problema industrial → datos → grafo → modelo estadístico → optimización → impacto.

Slide	Tema	Tiempo
1	Título y mensaje principal	1 min
2	Motivación industrial: OEE y changeovers	1.5 min
3	Datasets operacionales	1.5 min
4	Construcción del grafo	1.5 min
5	Dataset estadístico y EDA	1.5 min
6	Method 1: GoF y Bayesian updating	2 min
7	Resultados Method 1	2 min
8	Method 2: permutation tests	2 min
9	Resultados Method 2	1.5 min
10	Metodología de optimización	2 min
11	GA + Optuna + grafo aprendido	1.5 min
12	Resultados de optimización	2 min
13	Comparación visual de schedules	1 min
14	Conclusiones	1 min

Slide 1 – Title

Contenido de la diapositiva.

- **Statistical Characterization and Optimization of Format Changeovers in DAMM's El Prat Bottling Lines**
- From statistical changeover modelling to production-sequencing optimisation.
- Jose Calatayud Mateu.
- MSc in Advanced Mathematics – Statistics.

Gráficos / elementos visuales. No es necesario poner gráfico. Si quieres hacerlo más visual, usa una imagen de fondo industrial suave o el logo de DAMM/UAB/UB. Mantén esta diapositiva limpia.

Guion oral. Good morning. In this presentation I will explain how I used one year of operational data from DAMM's El Prat plant to model format changeovers statistically and then use that model to optimise a weekly production schedule.

The key idea is that a changeover is not just downtime. It depends on the sequence: producing one SKU after another can be easy or expensive depending on brand, volume, packaging and label reference. Therefore, if we learn which transitions are costly, we can build better production plans.

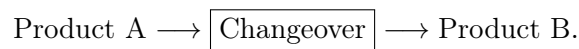
Slide 2 – Industrial Motivation: OEE And Changeovers

Contenido de la diapositiva.

$$\text{OEE} = \underbrace{\frac{\text{run time}}{\text{planned time}}}_{\text{Availability}} \times \underbrace{\frac{\text{ideal cycle} \times \text{units produced}}{\text{run time}}}_{\text{Performance}} \times \underbrace{\frac{\text{good units}}{\text{units produced}}}_{\text{Quality}}.$$

- DAMM operates three canning lines: L14, L17 and L19.
- Each line produces many SKUs.
- SKUs differ in brand, volume, pack and label reference.
- Changeovers mainly affect the **Availability** term.

Gráficos / elementos visuales. Pon una figura simple tipo:



No hace falta usar una figura externa aquí. Si quieres usar imagen, puedes usar la imagen de proceso de DAMM como contexto, pero sin cargar demasiado la slide.

Guion oral. The industrial KPI used in the project is OEE, Overall Equipment Effectiveness. It decomposes production losses into availability, performance and quality.

In this work, the relevant component is availability, because during a format change the line is stopped and no saleable output is produced. This loss is partly controllable because it depends on the sequence of products.

So the industrial question is: can we reduce avoidable downtime by choosing a better order of SKUs?

Slide 3 – Raw Operational Datasets

Contenido de la diapositiva.

Dataset	Role
OEE	efficiency indicators and product attributes
Cambios	observed changeover duration
Tiempo	time-accounting decomposition
Volumen	produced hL and units
Mantenimiento	maintenance contamination

Integration key = OF, manufacturing order.

Gráficos / elementos visuales. Recomendado: una versión simplificada del mapa de datasets que te pasó DAMM, o una red sencilla con **OF** en el centro y los cinco datasets alrededor.

Guion oral. The data came from five operational Excel exports. They are not independent tables; they are different views of the same manufacturing orders.

The key variable is the OF, the manufacturing order. It connects the product, the line, the OEE record, the produced volume, the time decomposition and the changeover information.

For the statistical model, the most important variable is **Frecuencia Total**, from the **Cambios** file, because it gives the observed changeover duration in hours.

Slide 4 – From Orders To A Directed Graph

Contenido de la diapositiva.

node = brand $\times$ volume $\times$ pack $\times$ label reference.

node = marca|volumen|pack|envase.

previous node $i \longrightarrow$ current node j = directed edge $i \rightarrow j$.

Edge type	Meaning
C_brand	brand changes
C_vol	volume changes
C_pack	pack changes
C_envase	label reference changes
C0_self	same complete node repeated

Gráficos / elementos visuales. Usa un dibujo de nodos y flechas. Si quieres usar una figura del proyecto, puedes poner `report/figures/problem_flow.png`, aunque esta slide también funciona muy bien con un esquema hecho en Canva.

Guion oral. A changeover cannot be defined from the current order alone. It is a transition: we need to know what was produced immediately before.

For that reason, each product configuration is converted into a graph node. The node combines brand, volume, pack and label reference.

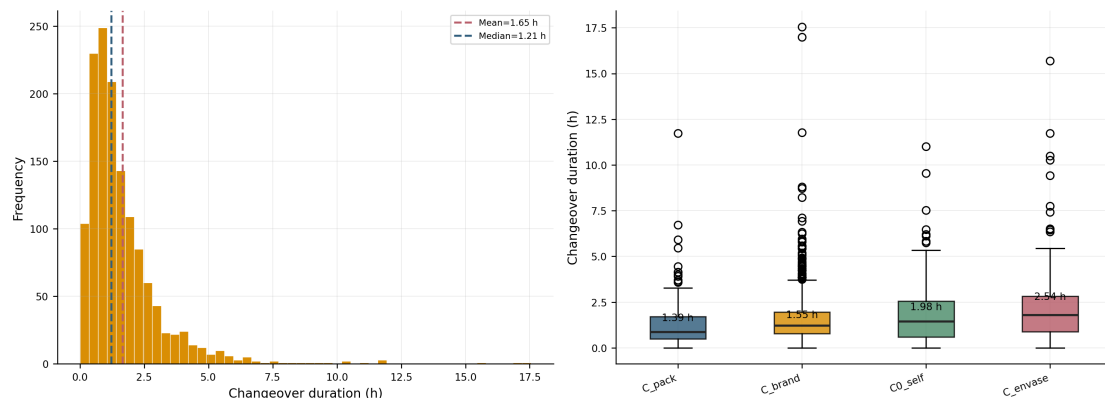
Then each changeover is a directed edge from the previous product node to the current product node. The edge type is defined using a priority rule: first brand, then volume, then pack, then label reference, and finally repeated format.

Slide 5 – Statistical Dataset And Exploratory Analysis

Contenido de la diapositiva.

- Post-mortem transition dataset:
 - 1,986 transitions.
 - 188 nodes.
 - 1,358 line-specific directed edges.
- Statistical duration dataset:
 - 1,378 observed changeover durations.
 - H_1 : 728 observations.
 - H_2 : 650 observations.
 - 184 nodes.
 - 1,015 unique directed edges.

Gráficos / elementos visuales. Usar obligatoriamente:



Archivo: `report/figures/nb07_fig01.png`.

Guion oral. There are two related datasets. The wider post-mortem dataset reconstructs all transitions between consecutive OFs. The stricter statistical dataset only keeps transitions with an observed duration.

The exploratory plot shows two important facts. First, durations are positive and strongly right-skewed. Second, the sample sizes by edge type are very unbalanced. **C_brand** dominates, while **C_vol** is rare.

This is why normal ANOVA or standard t-tests are not appropriate. The analysis needs positive-skew distributions and rank-based methods.

Slide 6 – Method 1: Goodness-Of-Fit And Bayesian Updating

Contenido de la diapositiva.

$$H_1 = \text{January--June 2025}, \quad H_2 = \text{July--December 2025}.$$

Candidate families: Gamma, Weibull, LogNormal, Inverse Gaussian.

$$\text{AIC} = -2\ell(\hat{\theta}) + 2k.$$

$$D = \sup_x |F_n(x) - F(x)|, \quad W^2 = \int [F_n(x) - F(x)]^2 dF(x).$$

Gráficos / elementos visuales. No pongas una figura compleja. Mejor un pipeline visual:

$$H_1 \rightarrow \text{fit distributions} \rightarrow \text{AIC} + \text{KS} + \text{CvM} \rightarrow \text{Bayesian cost}.$$

Si quieres poner figura real, usa `report/figures/nb07_fig02.png`, pero mejor reservar esa para hablar de resultados.

Guion oral. The first statistical method estimates the expected duration of a changeover.

The year is split into two halves. H_1 is used to fit the initial models and priors, while H_2 is used for validation and sequential updating.

Four positive-skew families are fitted by maximum likelihood. The comparison is based on AIC, but also on KS and Cramér–von Mises goodness-of-fit diagnostics. The final goal is not only to describe the distribution, but to obtain a stable cost that can be used by the optimiser.

Slide 7 – Bayesian Updating Layer

Contenido de la diapositiva.

$$X_i \mid \lambda \sim \text{Exponential}(\lambda), \quad \lambda \sim \text{Gamma}(a_0, b_0).$$

$$p(\mathbf{x} \mid \lambda) = \lambda^n e^{-\lambda \sum_i x_i}.$$

$$\lambda \mid \mathbf{x} \sim \text{Gamma} \left(a_0 + n, b_0 + \sum_i x_i \right).$$

$$\mathbb{E}[X \mid \mathbf{x}] = \mathbb{E} \left[\frac{1}{\lambda} \mid \mathbf{x} \right] = \frac{b_0 + \sum_i x_i}{a_0 + n - 1}.$$

Gráficos / elementos visuales. Pon una figura conceptual de actualización:

$$\text{prior} + \text{new observations} \rightarrow \text{posterior cost}.$$

No necesitas imagen externa en esta slide.

Guion oral. The Bayesian layer uses a Gamma–Exponential conjugate model.

Here, λ is the rate of the exponential distribution, and $1/\lambda$ is the expected duration.

The advantage of this model is that the posterior has a closed form. When new observations arrive, we only update the number of observations and the sum of durations.

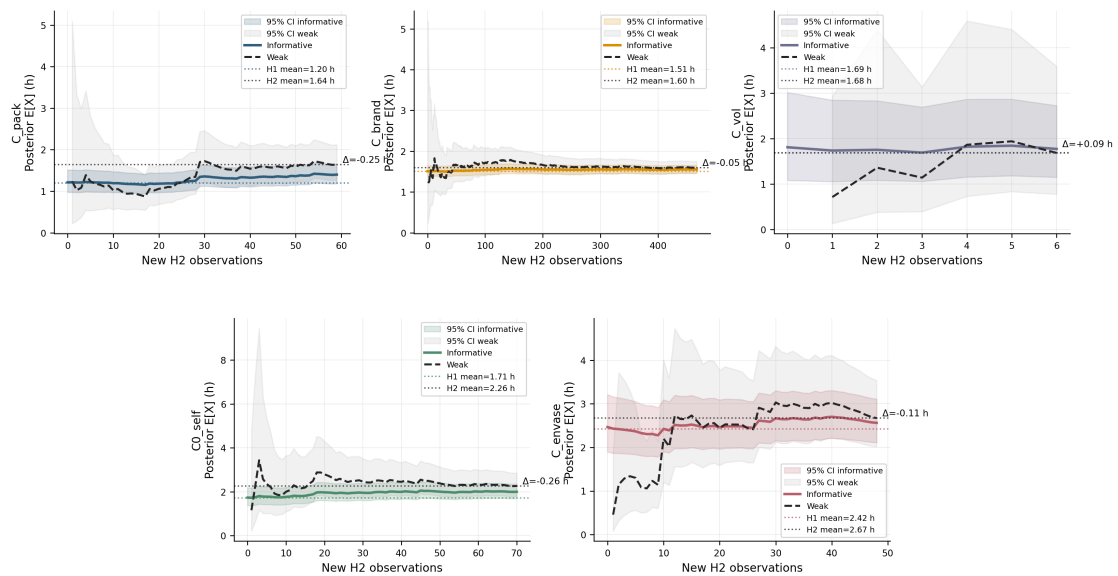
I compare two priors. The informative prior uses H_1 as historical pseudo-data. The weak prior starts almost without information and learns from H_2 .

Slide 8 – Method 1 Results

Contenido de la diapositiva.

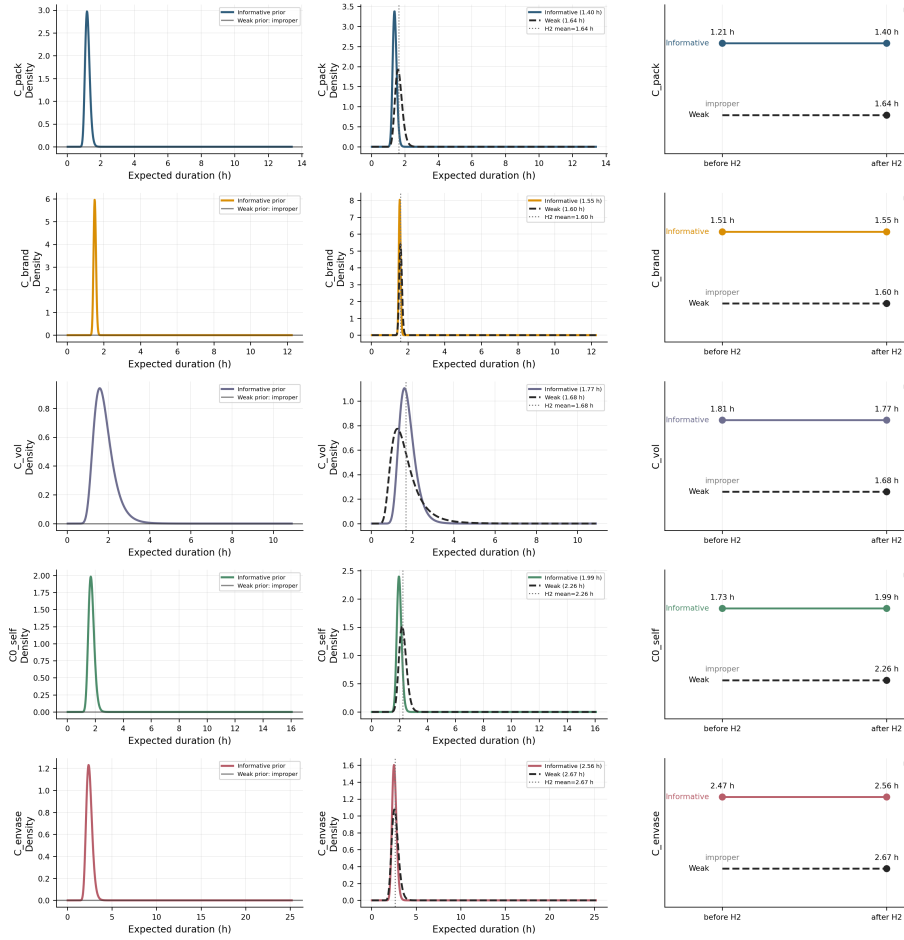
Type	Posterior mean
C_pack	1.399 h
C_brand	1.552 h
C_vol	1.772 h
C0_self	1.995 h
C_envase	2.561 h

Gráficos / elementos visuales. Gráfico principal:



Archivo: report/figures/nb07_fig03.png.

Gráfico opcional si tienes tiempo:



Archivo: `report/figures/nb07_fig04.png`.

Guion oral. The Gamma model is accepted for most edge types. For **C_brand**, all families are rejected because the sample is very large, so even small deviations are detected. However, Gamma still has the lowest AIC and is kept as a fallback.

The posterior means reveal an important operational pattern: **C_envase**, the label-reference change, has the largest expected duration. This is one of the main findings of the project.

The posterior plots show that informative and weak priors become close after observing H_2 , meaning that the final estimate is mainly driven by the data.

Slide 9 – Method 2: Ordered Permutation Tests

Contenido de la diapositiva.

$$C_{\text{pack}} \leq C_{\text{brand}} \leq C_{0_{\text{self}}} \leq C_{\text{envase}}.$$

$$L17 \leq L19 \leq L14.$$

$$R_i = \frac{1}{n_i} \sum_{j=1}^{n_i} r_{ij}.$$

$$T_k = \sum_{i < j} (R_j - R_i).$$

$$p = \frac{\# \{T^{(b)} \geq T_{\text{obs}}\} + 1}{B + 1}.$$

Gráficos / elementos visuales. No pongas demasiadas imágenes aquí. Esta slide es metodológica: usa las fórmulas y una flecha “rank data → permute labels → p-value”.

Guion oral. The second method tests ordered operational hypotheses.

Instead of comparing means with ANOVA, I rank all observations jointly. This is more robust for skewed and unbalanced samples.

If the ordered alternative is true, later groups should have larger mean ranks. The statistic T_k sums all ordered rank differences.

The p-value is obtained by permutation: under the null hypothesis, group labels are exchangeable, so I randomly shuffle the labels 10,000 times.

Slide 10 – Method 2 Results

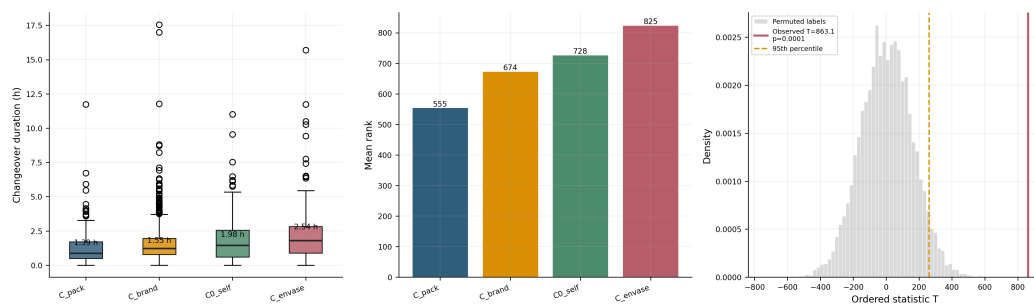
Contenido de la diapositiva.

$$T_4 = 863.1, \quad p = 10^{-4}.$$

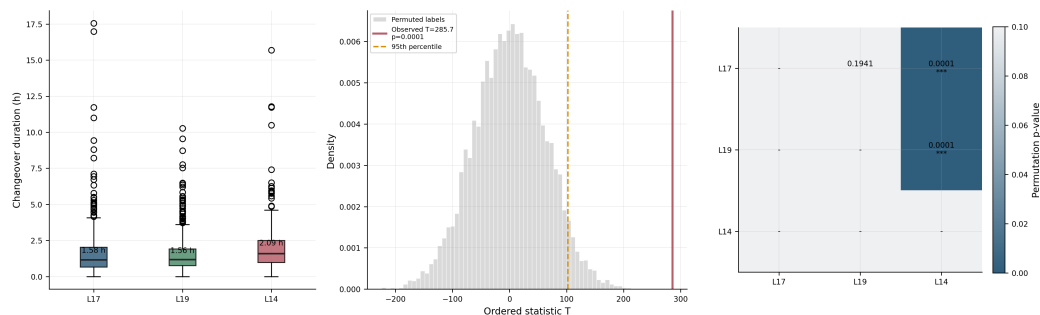
$$T_3 = 285.7, \quad p = 10^{-4}.$$

- C_pack is the lowest-cost group.
- C_envase is the highest-cost group.
- L14 is significantly slower than L17 and L19.
- L17 and L19 are not significantly different.

Gráficos / elementos visuales. Usa estos dos:



Archivo: `report/figures/nb07_fig07.png`.



Archivo: `report/figures/nb07_fig09.png`.

Opcional para post-hoc de edge types: `report/figures/nb07_fig08.png`.

Guion oral. The edge-type ordered test strongly rejects equality and supports the expected difficulty pattern.

The post-hoc analysis shows that the strongest differences involve C_pack at the low end and C_envase at the high end.

For lines, the ordered test also rejects equality. The post-hoc results show that the effect is mainly caused by L14, which is significantly slower than both L17 and L19.

Slide 11 – Optimisation Problem

Contenido de la diapositiva.

- Target week: 18–22 May 2026.
- Demand: 28 SKUs and 36,933.48 hL.
- Lines: L14, L17 and L19.
- Total weekly capacity: 340 h.

$$H_\ell = s_\ell + \sum_{i \in S_\ell} \frac{V_i}{\rho_{i\ell}} + \sum_{(i,j) \in A_\ell} c_{\ell ij}^h.$$

$$\min \sum_{\ell \in \{14,17,19\}} H_\ell, \quad H_\ell \leq \kappa_\ell.$$

Gráficos / elementos visuales. Usa un esquema en Canva con tres cajas:

Demand + throughput → assignment + sequencing → weekly hours.

No hace falta figura externa.

Guion oral. The optimisation problem is the weekly assignment and sequencing of the target week.

Each SKU must be assigned to one feasible line, and the SKUs assigned to a line must be ordered.

The total hours of a line have three components: start-up time, production time and changeover time.

The key term is $c_{\ell ij}^h$, the Bayesian expected changeover duration for the transition from SKU i to SKU j on line ℓ .

Slide 12 – Genetic Algorithm And Optuna

Contenido de la diapositiva.

$$\{14 : [\dots], 17 : [\dots], 19 : [\dots]\}.$$

$$\text{fitness} = \sum_{\ell} H_{\ell} + P_{\text{incompatibility}} + P_{\text{capacity}} + P_{\text{urgency}}.$$

$$\theta_{\text{GA}} = (\text{mut_prob}, \text{elitism}, \text{tour_size}).$$

Hyperparameter	Value
Mutation probability	0.849
Elitism	3
Tournament size	3
Best fitness	266.45

Gráficos / elementos visuales. Puedes usar una mini figura de pipeline:

$$\text{Optuna} \rightarrow \text{GA hyperparameters} \rightarrow \text{best schedule}.$$

Si quieres imagen real de Optuna, usa `report/figures/sec06_ga_optuna_result.png`, pero en la presentación es mejor una tabla limpia.

Guion oral. The final optimiser is a genetic algorithm.

A schedule is represented as a chromosome with one sequence per line. Every SKU appears exactly once across the three lists.

The fitness is total simulated hours plus penalties. The penalties ensure that the solution respects physical compatibility, historical eligibility, capacity and urgent orders.

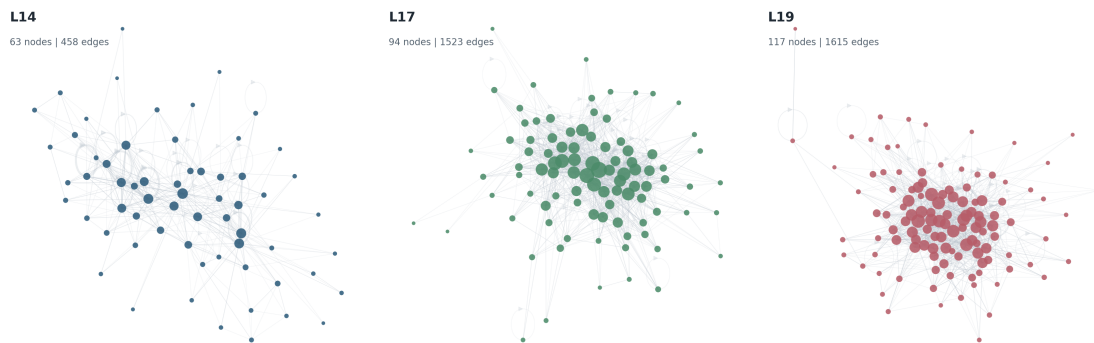
Optuna is used to tune the genetic algorithm. It searches over mutation probability, elitism and tournament size using a TPE sampler.

Slide 13 – Learned Bayesian Graph

Contenido de la diapositiva.

Quantity	Value
2025 production orders	2141
Product nodes	188
Unique directed edges	3596

Gráficos / elementos visuales. Usar:



Archivo: `report/figures/sec06_learned_graph_by_line.png`.

Guion oral. The Streamlit app learns the historical Bayesian changeover graph from the 2025 production history.

The graph contains 188 product nodes and 3,596 unique directed edges.

The graph is visualised separately by line because the same product transition can have different behaviour depending on whether it occurs on L14, L17 or L19.

This is the operational cost map used by the optimiser.

Slide 14 – Optimisation Results

Contenido de la diapositiva.

Schedule	Total hours
Real executed schedule	426.1 h
Theoretical planner schedule	333.0 h
GA optimised schedule	325.6 h

$$\text{Saving vs real} = 426.1 - 325.6 = 100.6 \text{ h.}$$

$$\text{Reduction} = 23.6\%, \quad 59.8\% \rightarrow 87.4\%, \quad \Delta\text{OEE} = +27.6 \text{ pp.}$$

Gráficos / elementos visuales. No pongas las tres schedules aquí si tienes la siguiente slide para ello. Aquí pon una tabla limpia y un número grande:

100.6 h saved

Guion oral. The final schedule is compared with two references: the real executed schedule and the theoretical planner schedule.

The real schedule requires 426.1 hours. The theoretical planner requires 333.0 hours. The GA optimised schedule requires 325.6 hours.

This means a saving of 100.6 hours compared with the real executed schedule, which is a 23.6 percent reduction.

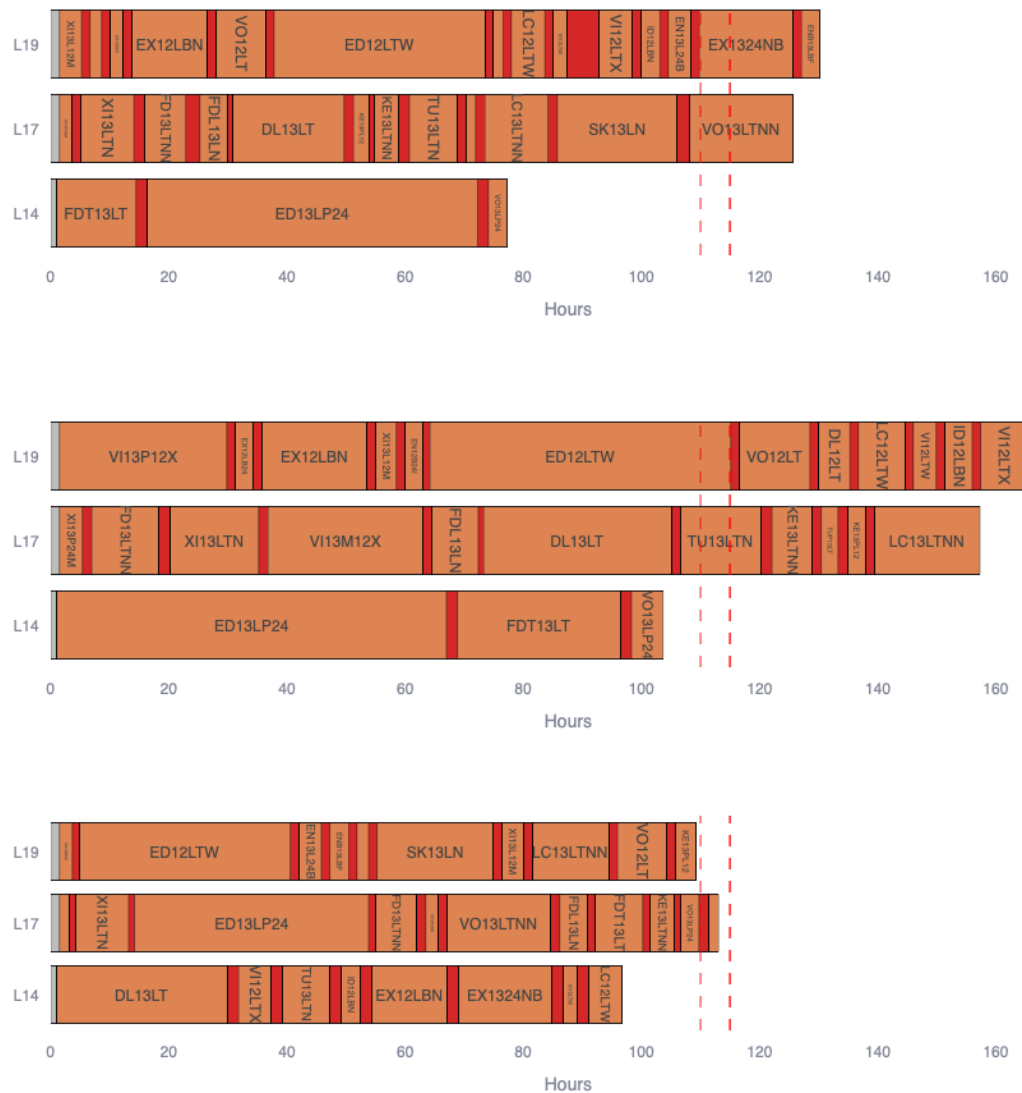
The OEE also improves substantially, from 59.8 percent to 87.4 percent.

Slide 15 – Schedule Comparison

Contenido de la diapositiva. Mostrar las tres planificaciones en este orden:

1. Theoretical planner.
2. Real executed.
3. GA optimised.

Gráficos / elementos visuales. Usar estas tres figuras:



Archivos:

- `report/figures/theoric_planner.png`
- `report/figures/newplot-3.png`
- `report/figures/optim_planner.png`

Guion oral. These three schedules show the difference visually.

The theoretical planner is already much better than the real executed schedule, but the GA optimiser still improves it.

The optimiser changes both assignment and sequencing. Some SKUs are moved to different lines, and the internal order of each line changes.

The purpose is not only to reduce individual changeovers, but to minimise the total weekly execution time while keeping the plan feasible.

Slide 16 – Conclusions

Contenido de la diapositiva.

- Changeovers are sequence-dependent and should be modelled as graph edges.
- Durations are positive, skewed and unbalanced.
- Gamma–Exponential Bayesian updating gives an operational cost layer.
- Ordered rank tests show:

$$C_{\text{pack}} \leq C_{\text{brand}} \leq C_{0_self} \leq C_{\text{envase}}.$$

- C_{envase} is a critical changeover type.
- The GA + Optuna optimiser saves 100.6 h in the target week.

Limitations

- One year of historical data.
- Some directed edges are sparse.
- Fallback rules are needed for unseen transitions.
- Future work: live validation with new production weeks.

Gráficos / elementos visuales. No hace falta gráfico. Puedes poner una frase final grande:

Changeovers are a statistical network problem, not only downtime.

Guion oral. To conclude, the project connects statistical modelling and production planning.

The statistical model is not only descriptive. It becomes the cost function of the optimiser.

The most actionable finding is that label-reference changes are expensive and should be considered explicitly in sequencing decisions.

Finally, the optimisation results show that using learned Bayesian changeover costs can significantly reduce weekly workload and improve OEE.

Resumen de gráficos que debes poner en Canva

Slide	Archivo	Uso
3	mapa DAMM o esquema propio	explicar cómo se conectan los datasets por OF
4	report/figures/problem_flow.png opcional	explicar nodos y edges
5	report/figures/nb07_fig01.png	EDA: distribución global y por edge type
8	report/figures/nb07_fig03.png	posterior sequential update
8	report/figures/nb07_fig04.png opcional	prior-to-posterior densities
10	report/figures/nb07_fig07.png	ordered edge-type test
10	report/figures/nb07_fig08.png opcional	post-hoc edge-type tests
10	report/figures/nb07_fig09.png	ordered line test
13	report/figures/sec06_learned_graph_by_line.png	learned Bayesian graph by line
15	report/figures/theoric_planner.png	theoretical planner schedule
15	report/figures/newplot-3.png	real executed schedule
15	report/figures/optim_planner.png	GA optimised schedule